Documentation

PF-Fatigue-GC: A Gradient-Consistent Phase-Field Fatigue Implementation for Abaqus (Version 1.0)

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Abstract

This document describes the Fortran subroutines, input files, scripts and Abaqus/CAE plug-in of PF-Fatigue-GC. The code simulates fatigue crack initiation and growth with the phase-field method in Abaqus/Standard. It runs on the built-in coupled temperature–displacement elements, and no user element is needed. If you use the code for research or industrial purposes, please cite:

- K. Kang, Y. Chen, Y. Liu. A Gradient-Consistent Abaqus UMAT/UMATHT Framework for Phase-Field Fatigue Fracture. *International Journal of Fatigue* (submitted).

The implementation builds on the heat-conduction analogy of Navidtehrani, Betegón and Martínez-Pañeda. Please also cite [1, 2]:

- Y. Navidtehrani, C. Betegón, E. Martínez-Pañeda. A unified Abaqus implementation of the phase field fracture method using only a user material subroutine. *Materials* 14 (2021) 1913.
- Y. Navidtehrani, C. Betegón, E. Martínez-Pañeda. A simple and robust Abaqus implementation of the phase field fracture method. *Applications in Engineering Science* 6 (2021) 100050.

Contents: 1 Introduction and list of files. 2 Abaqus implementation. 3 Usage. 4 Representative example. 5 Mode-II example. 6 Parameter identification. 7 Numerical settings and troubleshooting. Appendix A Formulation. Appendix B Heat-conduction analogy with a variable flux. Appendix C Cycle-domain fatigue law. Appendix D PF-Fatigue plug-in.

1. Introduction and list of files

In the phase-field method a crack is described by a smooth field ϕ, equal to 0 in intact material and 1 in fully broken material [3, 4]. The crack path follows from the field equations, so no criterion for the crack direction is needed. In fatigue models the fracture resistance decreases as a cyclic history accumulates [5, 6].

In Abaqus, the phase-field equation can be solved with the temperature degree of freedom of the coupled temperature–displacement elements, because the steady heat equation has the same form [1, 2]. In the published implementations the conductivity is one and the phase-field source is defined in UMAT (RPL) or in HETVAL. With a fatigue degradation function f, the fracture resistance $f\,G_c$ varies in space and the gradient term of the phase-field equation contains f as well. A constant conductivity cannot represent this term.

PF-Fatigue-GC defines the heat flux in a UMATHT subroutine, $\boldsymbol{q} = -f\nabla\phi$. The element assembly then contains the variable-coefficient gradient term, including $(\nabla \ln f) \cdot \nabla\phi$.

This form is called gradient-consistent (GC). The local-equivalent (LE) form keeps a unit conductivity and puts f only into the source. It is included for comparison (Appendix A.3).

The package contains two fatigue laws:

- **Cycle by cycle**: every load cycle is resolved with a fixed number of increments per cycle (IPC). It is used for benchmark problems with tens to hundreds of cycles.
- **Cycle domain (Envelope)**: the peak load is held and each increment represents ΔN cycles. It is used for long lives. Its three fatigue parameters can be determined from S–N or Paris data (Section 6).

Table 1 lists the files. All paths are relative to the root of the package.

Table 1: Files of the package

File and folder	Content
`src/PF_Fatigue_Exact_EnergySplit.f`	UMAT + UMATHT, cycle by cycle, GC form (7 constants)
`src/PF_Fatigue_LE_EnergySplit.f`	UMAT + UMATHT, cycle by cycle, LE form, for comparison
`src/PF_Fatigue_GC_Envelope_Final.f`	UMAT + UMATHT, cycle domain, GC form (9 constants)
`examples/SENT_I_tension/SENT_I_IPC128.inp`	example of Section 4
`examples/SENT_II_shear/SENT_II_IPC16.inp`	example of Section 5
`postprocessing/abaqus_python/`	crack length, crack path and crack-tip history from the ODB
`postprocessing/identification/`	fatigue parameters from S–N and Paris data, with the experimental data of the article
`plugin/PF_Fatigue/`	Abaqus/CAE plug-in PF-Fatigue v1.0 (Appendix D)
`manual/PF-Fatigue_v1.0_User_Guide.pdf`	step-by-step guide to the plug-in
`README.md`	overview, quick start and file list

The code was tested with Abaqus/Standard 2023 and Intel Fortran 19.1 (Visual Studio 2019) on Windows. RPL in UMAT for a coupled temperature–displacement analysis requires Abaqus 2020 or newer [2].

2. Abaqus implementation

2.1 Tasks of the subroutines

The temperature degree of freedom T is the phase field ϕ. The crack is seen by plotting the nodal temperature NT11. At each integration point Abaqus passes the strain, the temperature and its gradient, the state variables (SDV) and the material constants (PROPS) to the subroutines (**Fig. 1**).

1. UMAT computes the undamaged stress $\boldsymbol{\sigma_0} = \boldsymbol{C_0}\colon\boldsymbol{\varepsilon}$ and degrades it with $g(\phi_n) = (1-\phi_n)^2 + k_{\mathrm{res}}$. It updates the history field $H = \max(H, \psi_0^+)$, the fatigue history and evaluates the degradation f. It returns the phase-field source r as RPL and $\partial r/\partial\phi$ as DRPLDT, and stores f in SDV4.
2. UMATHT reads f from SDV4 and returns the flux $\boldsymbol{q} = -f\nabla\phi$ (FLUX) and its derivative $-f\boldsymbol{I}$ (DFDG). The gradient-consistent term enters only here.
3. Abaqus assembles the displacement and phase-field equations.

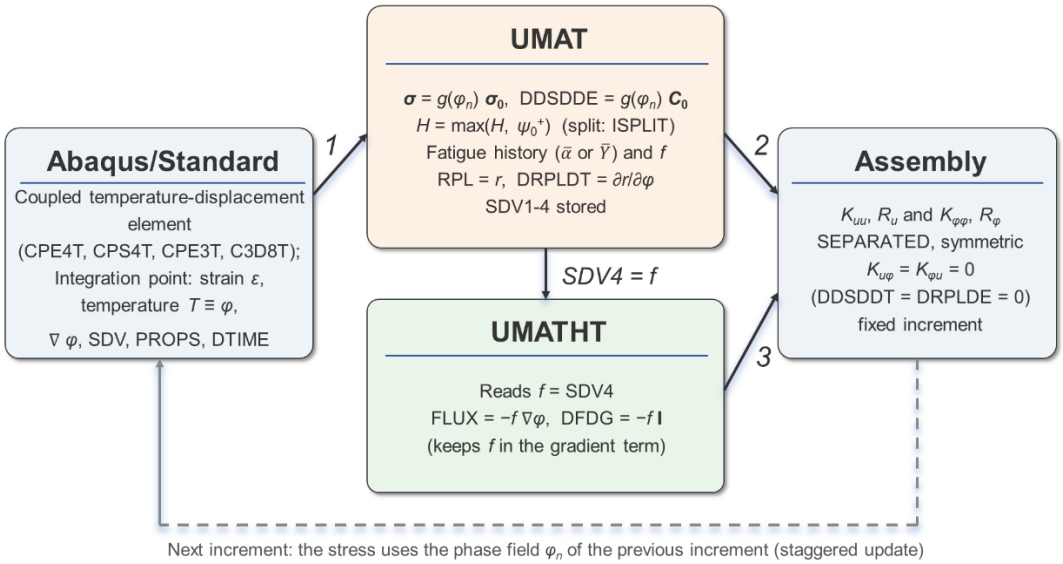


Fig. 1. Flowchart of the user subroutines. UMAT supplies the stress, the history variables and the phase-field source. UMATHT supplies the variable flux. The two fields are solved separately and the stress uses the phase field of the previous increment.

The source and the flux are (Appendix B)

$$r = \frac{2(1-\phi)H}{G_c l_0} - \frac{f\phi}{l_0^2} \tag{1}$$

$$\frac{\partial r}{\partial \phi} = -\frac{2H}{G_c l_0} - \frac{f}{l_0^2} \tag{2}$$

$$\boldsymbol{q} = -f\nabla\phi \tag{3}$$

$$\frac{\partial \boldsymbol{q}}{\partial(\nabla\phi)} = -f\,\boldsymbol{I} \tag{4}$$

2.2 Solution scheme

The coupling terms between displacement and phase field are not defined (DDSDDT = DRPLDE = 0). The stiffness matrix is symmetric, and the two fields are solved with the separated solution technique. The stress is computed with the phase field at the start of the increment, ϕ_n, and the fatigue history is updated once per increment from converged values. This is the single-pass staggered scheme of [2]; the result depends on the increment size. Fixed increments are used, and their size is checked by refinement (Section 7).

2.3 Constitutive model

- Small strain, isotropic linear elasticity, AT2 phase field (Appendix A).
- Energy split of the crack driving force, selected by ISPLIT: 0 no split, 1 volumetric–deviatoric (Amor et al. [7]), 2 spectral (Miehe et al. [8]), 3 no-tension. The split acts only on the driving force. The whole stress is degraded by $g(\phi)$ (hybrid formulation [9]).
- Fatigue degradation of Carrara et al. [5]: $f = 1$ below a threshold and $f = (2\alpha_T/(\bar{\alpha} + \alpha_T))^2$ above it (Appendix A.2), with a lower bound of 10^{-12}.

3. Usage

A model is built as an ordinary Abaqus model with the additions below. The plug-in of Appendix D performs Sections 3.2 to 3.7. This section is for users who build the model by hand or edit an input file.

3.1 Subroutine

Table 2: Choice of the subroutine

Task	File	Constants	Step time
resolved cycles (benchmarks, short tests)	`PF_Fatigue_Exact_EnergySplit.f`	7	cycles
long lives, S–N, Paris, structures	`PF_Fatigue_GC_Envelope_Final.f`	9	cycles
comparison with the LE form	`PF_Fatigue_LE_EnergySplit.f`	7	cycles

The cycle-by-cycle and the cycle-domain files use different constants and state variables. They cannot be exchanged without changing the material definition.

3.2 Material

In the Property module, create the material (Material Manager → Create):

1. General → Depvar: number of solution-dependent state variables = 4.
2. General → User Material: mechanical and thermal constants (type THERMOMECHANICAL in the scripting interface).
    - Mechanical constants: Table 3 (7 or 9 values).
    - Thermal constants: one value, 1.
3. General → Density: 1, as in the example files. It has no effect in a steady-state analysis.
4. No elastic, conductivity or specific-heat data. UMAT supplies the stiffness and UMATHT the flux.

The constants are given in the input file, and the Fortran file is not edited.

Table 3: Material constants (PROPS)

PROPS	Cycle by cycle (`Exact`, `LE`)	Cycle domain (`Envelope_Final`)
1	Young's modulus E	E
2	Poisson's ratio ν	ν
3	fracture energy G_c	G_c
4	length scale l_0	l_0
5	fatigue threshold α_T (energy density)	dimensionless threshold η_T
6	residual stiffness k_{res}	ISPLIT
7	ISPLIT	exponent p
8	–	stress ratio R (< 1)
9	–	mean-stress parameter κ

In the cycle-domain file $k_{\mathrm{res}} = 10^{-6}$ is fixed in the code. The cycle-by-cycle threshold $\alpha_T = G_c/(12\, l_0)$ is the value of the benchmark of [10].

Table 4: Solution-dependent state variables (SDV)

SDV	Cycle by cycle	Cycle domain
1	accumulated fatigue variable $\bar{\alpha}$	accumulated history $\bar{Y}$
2	history field H	history field H
3	α of the previous increment	equivalent driving force Y_{eq}
4	degradation f	degradation f

UMATHT reads SDV4. It must not be used for other purposes. In the source code, $\bar{Y}$, Y_{eq}, $Y_{\max}$ and β of Appendix C are named Abar, Aeq, Amax and S.

Assign the material through a solid section (thickness 1 for plane problems, unless the real thickness is wanted).

3.3 Step

Create a Coupled temp-displacement step:

1. Basic tab: Response = Steady-state; Nlgeom off; time period = number of cycles (Section 3.4).
2. Incrementation tab: Type = Automatic, with Initial = Minimum = Maximum increment size = Δt, and a maximum number of increments larger than (time period)/Δt.
3. Other tab: Solution technique = Separated; Equation solver matrix storage = Symmetric; creep/viscoelastic integration: none.

Table 5: Increment of the two fatigue laws

Fatigue law	Increment Δt	Meaning
cycle by cycle	$1/\mathrm{IPC}$, IPC a multiple of 4	one unit of step time is one load cycle
cycle domain	ΔN	one increment advances the fatigue history by ΔN cycles

The example files run with the default solution controls. If an analysis stops because the maximum number of equilibrium iterations is reached, the time-incrementation controls can be relaxed as described in [2].

3.4 Loads and amplitude

Apply the supports and one driving load or displacement as in a static model, and give it an amplitude:

- Cycle by cycle: a tabular triangular amplitude with one cycle per unit of step time. For the fully reversed SENT example the points are $(0,0)$, $(0.25,1)$, $(0.75,-1)$, $(1,0)$, $(1.25,1)$, … For a stress ratio R the plug-in uses a quarter-cycle ramp to the peak and then cycles between 1 and R. The step time is then the number of cycles + 0.25.
- Cycle domain: a constant amplitude equal to 1, so the peak load is held. The stress ratio enters only through PROPS(8). A cyclic amplitude must not be used, because each increment already represents ΔN cycles.

3.5 Initial crack

Two options:

- Geometric crack: a seam or a notch in the mesh. Both example files use this option.
- Phase-field crack: $\phi = 1$ is held on a node set with a temperature boundary condition in the fatigue step, with a constant amplitude equal to 1.

3.6 Mesh

- Element family: coupled temperature–displacement. Tested element types: CPE4T, CPS4T, CPE3T (with CPE4T) and C3D8T.
- Element size on the crack path: $h \leq l_0/2$. The SENT examples use $h = l_0/4$.

3.7 Output

Request field output of U, RF, S, NT and SDV. The SENT-I example writes field output every cycle (frequency = IPC = 128), the SENT-II example every 10 cycles (frequency = 160 with IPC = 16). Table 6 lists the variables to plot in the Visualization module.

Table 6: Field output

Output	Meaning
NT11	phase field ϕ
SDV1	accumulated fatigue variable ($\bar{\alpha}$ or $\bar{Y}$)
SDV2	history field H
SDV4	fatigue degradation f

Crack length and crack path can be extracted with the scripts in `postprocessing/abaqus_python`.

3.8 Running the job

From a command window in the folder of the input file:

```
abaqus job=SENT_I_GC input=SENT_I_IPC128.inp user=..\..\src\PF_Fatigue_Exact_Energy
Split.f cpus=8 interactive
```

In Abaqus/CAE the Fortran file is given in Job → Edit → General → User subroutine file. The Fortran compiler must be linked to Abaqus; `abaqus verify -user_std` checks this.

3.9 Keywords

The material and step definitions of the cycle-by-cycle example are

```
*Material, name=Material-1
*Density
1.
*Depvar
4,
*User Material, type=MECHANICAL, constants=7
210000., 0.3, 2.7, 0.004, 56.25, 1.e-7, 0.
*User Material, type=THERMAL, constants=1
1.
*Step, name=Step-1, nlgeom=NO, inc=400000, unsymm=NO
*Coupled Temperature-displacement, creep=none, steady state
0.0078125, 350., 0.0078125, 0.0078125
*Solution Technique, type=SEPARATED
```

For the cycle-domain file the corresponding lines are (symbols to be replaced by numbers)

```
*User Material, type=MECHANICAL, constants=9
E, nu, Gc, l0, eta_T, ISPLIT, p, R,
kappa
*User Material, type=THERMAL, constants=1
1.
*Step, name=Step-1, nlgeom=NO, amplitude=STEP, inc=10000000, unsymm=NO
*Coupled Temperature-displacement, creep=none, steady state
dN, N_total, dN, dN
*Solution Technique, type=SEPARATED
```

with the peak load applied with a constant amplitude equal to 1.

4. Representative example

The single-edge-notched tension (SENT) specimen of Kristensen and Martínez-Pañeda [10] is a common benchmark for phase-field fatigue. A 1 × 1 mm plate contains a horizontal edge crack of length 0.5 mm (**Fig. 2**(a)). The bottom edge is fixed vertically, and the top edge is displaced by a fully reversed triangular history with an amplitude of 0.002 mm (**Fig. 2**(b)).

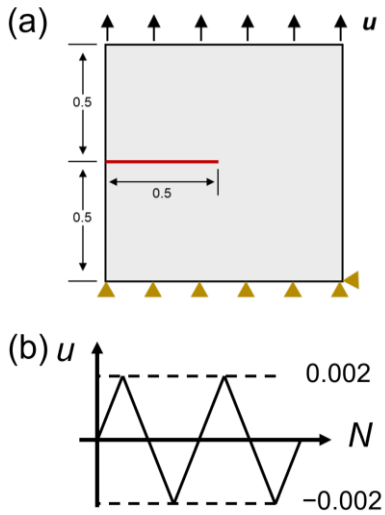


Fig. 2. SENT specimen. (a) Geometry and boundary conditions, dimensions in mm. (b) Applied displacement history.

The material constants are $E = 210$ GPa, $\nu = 0.3$, $G_c = 2.7$ N/mm and $l_0 = 0.004$ mm, with $\alpha_T = G_c/(12l_0) = 56.25$ MPa, $k_{\text{res}} = 10^{-7}$ and no energy split (ISPLIT = 0). The mesh has 56,737 CPE4T elements, with $h = 0.001$ mm $= l_0/4$ on the crack path. Each cycle is resolved with 128 fixed increments (IPC = 128), and 350 cycles are requested. The input file is `examples/SENT_I_tension/SENT_I_IPC128.inp`. The crack extension a is the distance from the initial tip to the farthest node with $\phi \geq 0.95$ (`extract_crack_curve_SENT.py`).

Fig. 3 compares the crack extension with the reference result. The GC file reaches a = 0.1, 0.2, 0.3 and 0.4 mm after 91, 144, 186 and 221 cycles, 2–5% earlier than the reference at the same IPC. The LE file, run with the same input file, reaches these crack lengths 14–25% earlier than the GC file. In both cases the crack stays straight. The crack reaches the right edge after 247 cycles (GC) and 211 cycles (LE).

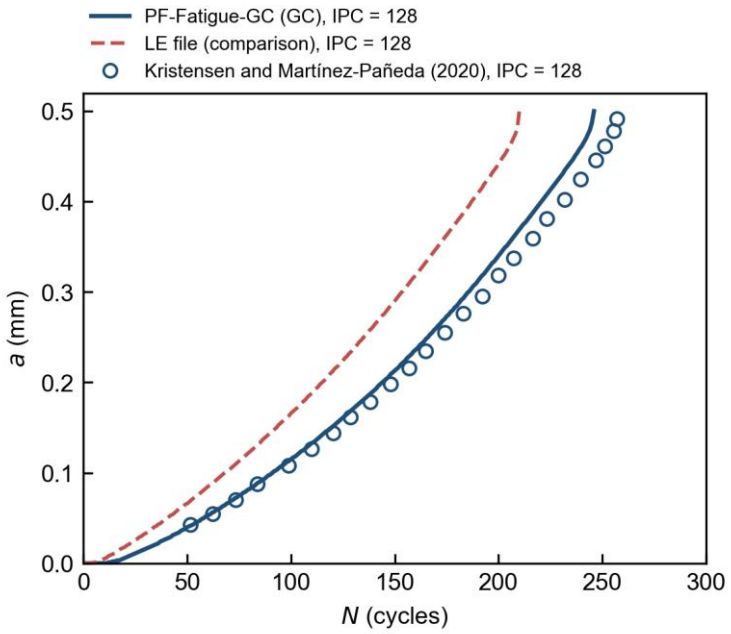


Fig. 3. Crack extension versus number of cycles for the SENT specimen at IPC = 128: GC and LE files of this package, and the result of Kristensen and Martínez-Pañeda [10] at the same IPC.

5. Mode-II example

`examples/SENT_II_shear/SENT_II_IPC16.inp` has the same geometry as the Mode-I example, with 268,518 CPE4T elements. The top edge is displaced horizontally from 0 to 0.006 mm in every cycle ($R = 0$), and the spectral split is used (ISPLIT = 2). Each cycle has 16 increments (IPC = 16), and 2000 cycles are requested. The crack kinks downward and follows a curved path. The same input file is run with the GC or the LE file.

The crack extension along the curved path is extracted with `extract_crack_curve_general.py`, the crack paths at matched crack length with `export_sent2_path_raw.py`, and the initiation angle with `extract_SENT2_initiation_angles.py`. All files use N, mm and MPa.

The input files of the other examples of the article are not part of this package.

6. Parameter identification for the cycle-domain law

The cycle-domain law has three fatigue parameters: the exponent p, the threshold η_T and the mean-stress parameter κ. Because of the cycle scaling of the law (Appendix C), p sets the slope of the S–N curve and the Paris exponent. η_T and κ rescale the cycle axis: η_T sets the level of the curves and κ the spacing between the curves of different stress ratios. Data at two stress ratios are enough.

From S–N data (`postprocessing/identification/sn_identify.py`). In a homogeneous specimen the life is

$$N_f = \eta_T \, \beta^{-2\kappa p} \, F_p\left(\frac{\sigma_{\max}}{\sigma_c}\right) \tag{5}$$

with $\beta = (1 - R)/2$ and the AT2 strength $\sigma_c = \sqrt{27 E G_c/(256\, l_0)}$ [11]. The script finds p from the slope of the curve at $R = 0$. The model slope increases with p, so the measured slope gives a single value of p. At low stress the model slope is $2p + 1$. η_T and κ then follow from the mean offsets of the curves at $R = -1$ and $R = 0$. l_0 is chosen so that σ_c equals the tensile strength. The script reads `sn_experimental_data.csv` and reproduces the parameters of the article.

From Paris data (`postprocessing/identification/paris_identify.py`):

1. Run a few reference calculations of the specimen with $\eta_T = 1$, $R = -1$ ($\beta = 1$), $\Delta N = 0.625$ and trial values of p. Extract the crack-tip histories with `extract_CT_crack_history.py`.
2. `python paris_identify.py identify <material> <folder of the histories>` removes the first $20\, l_0$ of crack extension, interpolates the reference rates in p to the measured Paris exponent, and computes η_T and κ from the offsets of the two calibration curves. Each measured curve is used in its Paris interval (article, Supporting Information S4). The script contains the CT geometry of the two materials of the article; for another specimen the entry `GEOMETRY` is edited.
3. Run the specimen with the identified values. For any stress ratio, $\Delta N = 0.625\, \eta_T / \beta^{2\kappa p}$ gives the same scaled increment $\Delta N^* = 0.625$.

S–N and Paris identifications use the same model and parameter definitions, but the values are fitted to different data.

7. Numerical settings and troubleshooting

Table 7: Numerical settings

Setting	Recommendation
element size on the crack path	$h \le l_0/2$
cycle-by-cycle resolution	results depend on IPC; compare models at the same IPC
cycle-domain resolution	set by $\Delta N^* = \Delta N\, \beta^{2\kappa p}/\eta_T$, not by ΔN; the CT calculations of the article use $\Delta N^* = 0.625$ (halving it changed the cycles to a given crack length by less than 3.5%); check by halving ΔN once
initial transient	growth rates in the first $\approx 20\, l_0$ from a sharp notch are not in the Paris regime
`.sta` file	the increment size must be constant

Table 8: Troubleshooting

Problem	Remedy
`LNK1105: cannot close file "standardU.dll"`	the linker could not finish (simultaneous compilations or a virus scanner); the solver has not started; submit the job again
`Problem during compilation`	link the Fortran compiler to Abaqus; `abaqus verify -user_std`
`NPROPS must be at least 7 / 9`	the constants do not match the subroutine (Table 3)
`NSTATV must be at least 4`	define `*Depvar` = 4
`stress ratio R must be < 1`	correct PROPS(8)
the initial crack is not visible	use a geometric crack or a temperature BC with constant amplitude (Section 3.5)
the crack grows by several elements in one increment	increase IPC or reduce ΔN
the increments grow during the step	set initial = minimum = maximum increment (Section 3.3)

Appendix A. Formulation

A.1 Phase field and history field

For a body Ω with displacement $\boldsymbol{u}$ and small strain $\boldsymbol{\varepsilon}$, the AT2 model uses [3, 4]

$$g(\phi) = (1-\phi)^2 + k_{\mathrm{res}} \quad (6)$$

$$\gamma(\phi, \nabla\phi) = \frac{\phi^2}{2l_0} + \frac{l_0}{2}|\nabla\phi|^2 \quad (7)$$

The stress is $\boldsymbol{\sigma} = g(\phi)\,\boldsymbol{\sigma_0}$. The tensile energy ψ_0^+, selected by ISPLIT, drives the phase field through the history field

$$H(\boldsymbol{x}, t) = \max_{\tau \le t} \psi_0^+\big(\boldsymbol{\varepsilon}(\boldsymbol{x}, \tau)\big) \quad (8)$$

H does not decrease on unloading, which approximates the irreversibility of the crack. The phase field itself is not constrained to increase.

A.2 Fatigue degradation

The fatigue degradation f multiplies the fracture energy density, $f\, G_c\, \gamma$ [5]. In the cycle-by-cycle law the fatigue variable accumulates the positive increments of $\alpha = g(\phi)\,\psi_0^+$:

$$\bar{\alpha}_{n+1} = \bar{\alpha}_n + \max(\alpha_{n+1} - \alpha_n, 0) \quad (9)$$

$$f(\bar{\alpha}) = \begin{cases} 1, & \bar{\alpha} \le \alpha_T, \\ \left(\dfrac{2\alpha_T}{\bar{\alpha} + \alpha_T}\right)^2, & \bar{\alpha} > \alpha_T. \end{cases} \quad (10)$$

The cycle-domain law is given in Appendix C.

A.3 Gradient-consistent and local-equivalent equations

With f fixed during the solution for ϕ, the weak form of the phase-field equation is

$$\int_\Omega \left[f G_c l_0 \nabla\phi \cdot \nabla\delta\phi + \left(\frac{f G_c}{l_0} + 2H \right) \phi\, \delta\phi - 2H\delta\phi \right] \mathrm{d}\Omega = 0 \tag{11}$$

Division by the constant $G_c l_0$ gives the strong form solved by the GC files,

$$\nabla \cdot (f \nabla\phi) + \frac{2(1-\phi)H}{G_c l_0} - \frac{f\phi}{l_0^2} = 0 \tag{12}$$

with $f\nabla\phi \cdot \boldsymbol{n} = 0$ on free boundaries. Expanding the divergence gives the term $(\nabla \ln f) \cdot \nabla\phi$, which the LE form omits. The LE file solves

$$\nabla^2\phi + \frac{2(1-\phi)H}{f G_c l_0} - \frac{\phi}{l_0^2} = 0 \tag{13}$$

The two forms are the same when f is uniform. In the examples of the article the LE form gives faster crack growth (**Fig. 3**) and almost the same crack path under Mode-II loading.

Appendix B. Heat-conduction analogy with a variable flux

The steady heat equation with flux $\boldsymbol{q}$ and internal heat generation r is

$$-\int_\Omega \boldsymbol{q} \cdot \nabla\delta T \,\mathrm{d}\Omega - \int_\Omega r\ \delta T \,\mathrm{d}\Omega = 0 \tag{14}$$

For a conductor with $\boldsymbol{q} = -k\nabla T$ it becomes $\int_\Omega k\, \nabla T \cdot \nabla\delta T \,\mathrm{d}\Omega = \int_\Omega r\ \delta T \,\mathrm{d}\Omega$. With $T \equiv \phi$ it is the phase-field equation of Appendix A.3 when the Abaqus variables of Table 9 are defined.

Table 9: Abaqus variables of the heat-conduction analogy

Abaqus variable	Subroutine	GC files	LE file
temperature NT11	–	ϕ	ϕ
FLUX	UMATHT	$-f\,\nabla\phi$	$-\nabla\phi$
DFDG	UMATHT	$-f\,I$	$-I$
U, DUDT, DUDG, DFDT	UMATHT	0 (steady state)	0
RPL	UMAT	$\frac{2(1-\phi)H}{G_c l_0} - \frac{f\phi}{l_0^2}$	$\frac{2(1-\phi)H}{f G_c l_0} - \frac{\phi}{l_0^2}$
DRPLDT	UMAT	$-\frac{2H}{G_c l_0} - \frac{f}{l_0^2}$	$-\frac{2H}{f G_c l_0} - \frac{1}{l_0^2}$
DDSDDT, DRPLDE	UMAT	0 (staggered)	0

Appendix C. Cycle-domain fatigue law and cycle scaling

The cycle-domain file advances the fatigue history in the cycle domain. With the reference energy $\alpha_{T0} = G_c/(12 l_0)$,

$$Y_{\max} = \frac{g(\phi_n)\,\psi_0^+}{\alpha_{T0}} \tag{15}$$

$$Y_{\mathrm{eq}} = Y_{\max}\beta^{2\kappa} \tag{16}$$

$$\beta = \frac{1-R}{2} \tag{17}$$

$$\bar{Y}_{N+\Delta N} = \bar{Y}_N + \Delta N\, Y_{\mathrm{eq}}^p \tag{18}$$

$$f(\bar{Y}) = \begin{cases} 1, & \bar{Y} \le \eta_T, \\ \left(\dfrac{2\eta_T}{\bar{Y} + \eta_T} \right)^2, & \bar{Y} > \eta_T, \end{cases} \tag{19}$$

where ψ_0^+ is evaluated at the peak load and ΔN is the increment of step time. The accumulation has the form of the law of Golahmar et al. [6] with the Walker factor for the stress ratio. The endurance-limit term of that law is omitted, the energy is normalized with α_{T0}, and one increment represents ΔN cycles.

Cycle scaling. Under a fixed peak load, R and κ enter only through the constant factor $\beta^{2\kappa p}$ of the accumulation rate, and η_T only through $\bar{Y}/\eta_T$. With $\bar{Y}^* = \bar{Y}/\eta_T$ and $N^* = \beta^{2\kappa p} N/\eta_T$ the equations become $\mathrm{d}\bar{Y}^*/\mathrm{d}N^* = Y_{\max}^{p}$ and $f = (2/(\bar{Y}^* + 1))^2$ for $\bar{Y}^* > 1$, without η_T, κ or R. Two analyses that start from the same state under the same peak load give the same fields at the same N^*, at different N. For $\eta_T = 1$ and $R = -1$ ($\beta = 1$), $N^* = N$, so an analysis with these values gives the common solution (the reference problem). A state that the reference problem reaches after N^* cycles, for example a given crack length, is reached after

$$N = \eta_T \, \beta^{-2\kappa p} \, N^* \tag{20}$$

cycles, and the growth rate is divided by the same factor:

$$\frac{\mathrm{d}a}{\mathrm{d}N} = \eta_T^{-1} \, \beta^{2\kappa p} \, \frac{\mathrm{d}a}{\mathrm{d}N^*} \tag{21}$$

The increment in the reference problem is

$$\Delta N^* = \frac{\Delta N \, \beta^{2\kappa p}}{\eta_T} \tag{22}$$

The identification of Section 6 and the choice of ΔN are based on these relations. When the reference life follows a power law $N^* \propto \sigma_{\max}^{-m_S}$, the law takes the Walker form with exponent $\gamma_W = 2\kappa p/m_S$ [12].

Appendix D. PF-Fatigue plug-in

The Abaqus/CAE plug-in PF-Fatigue v1.0 (`plugin/PF_Fatigue`) performs Sections 3.2 to 3.7 from a dialog. The user builds the part, the mesh, the supports and one driving load or boundary condition. The plug-in then

- creates the material with the constants of Table 3 and the state variables, and assigns it (it creates a section if the model has none);
- creates or updates the coupled temperature–displacement step with fixed increments, the separated solution technique and symmetric storage;
- creates the triangular (cycle by cycle) or constant (cycle domain) amplitude and applies it to the driving load;
- requests the field output of Section 3.7;
- holds $\phi = 1$ on an initial-crack set if requested (Section 3.5);
- creates the job with the bundled subroutine and checks its checksum.

Check Model lists every check as `[OK]`, `[WARN]` or `[ERROR]`: element types, mesh size relative to l_0, stress ratio, IPC and the scaled increment ΔN^*. It warns when $\Delta N^* > 1$.

To install the plug-in, close Abaqus/CAE and copy the folder `PF_Fatigue` into a plug-in directory, for example `abaqus_plugins` in the home directory or the `plugins` folder of the Abaqus installation. Older versions must be removed from that directory. The menu entry

Plug-ins → PF-Fatigue appears in a new CAE session (Fig. 4). The User Guide `manual/PF-Fatigue_v1.0_User_Guide.pdf` describes every step with screenshots.

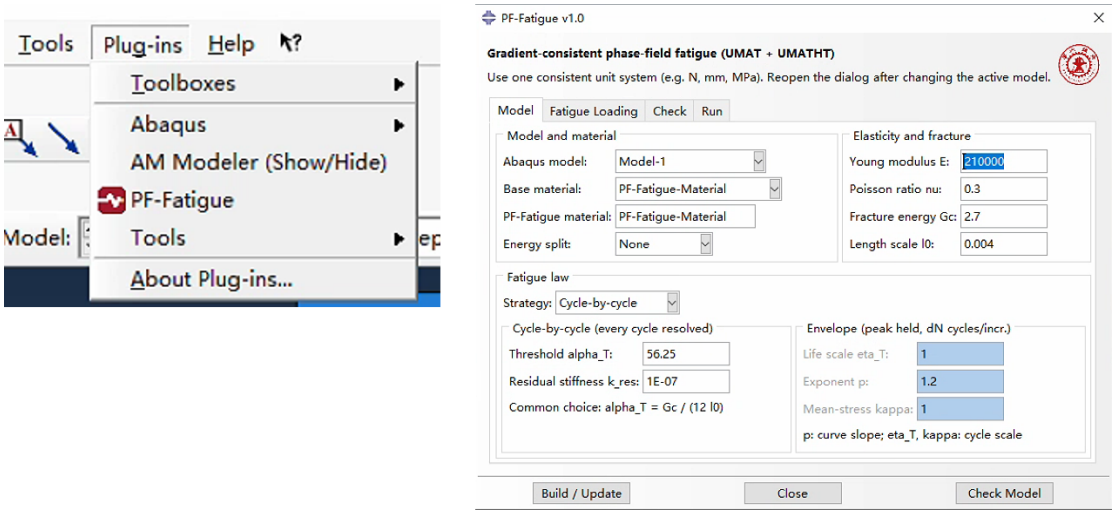


Fig. 4. Plug-ins menu with the PF-Fatigue entry (left) and the Model tab of the dialog (right).

The plug-in covers small strain, isotropic linear elasticity, constant-amplitude loading, one driving load and the element types of Section 3.6. The LE file is not available in the plug-in.

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